

Equivalence-class hierarchy for DR-recoverable templates

DR-recoverable templates

$$|V_{3D}^*| = 24 \times 21 \times 9! = 182,891,520$$



quotient by G , $|G| = 1,296$
every stabiliser is trivial

Equivalence classes (orbits under G)

$$182,891,520 / 1,296 = 141,120$$

Role level

24 Latin cubes of order 3,
a single orbit under G
(stabiliser of order 54)

Parity level

21 partitions of the cube into
nine parity transversals,
each labelled in $9!$ ways

$24 \times 21 \times 9!$ factors the count; the two levels are independent conditions